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Quantum kernel learning for breast cancer classification

Quantum Kernel Learning for Breast Cancer Classification: Advances, Benchmarks, and Future Directions (2023–2026)

Quick Reference

Key Findings Table

Method/Model	Dataset(s) Used	Recall (%)	F1 (%)	ROC-AUC	Computational Trade-off	Feasibility (48h, Qiskit/PennyLane)	DOI/Reference
QSVM (ZZ-Feature Map)	Wisconsin Diagnostic	70–94	70–94	0.727–0.995	Higher runtime, limited scalability	Yes (simulator)	1 2
Trainable Quantum Kernels	Wisconsin Diagnostic, METABRIC	90–98	90–98	0.98–0.998	Sub-sampling reduces cost	Yes (simulator)	3 4
Hybrid QSVM-QNN	Wisconsin Diagnostic, BreastMNIST	90–98	90–98	0.98–0.995	Reduced cost, improved accuracy	Yes (simulator)	4 5
Classical SVM (RBF)	Wisconsin Diagnostic, METABRIC	94–98	94–98	0.98–0.998	Fast, scalable	Yes	2 6
Custom Quantum Feature Maps	Wisconsin Diagnostic, BreastMNIST	+5% (AUC), +18% (F1)	+18% (F1)	+5% (AUC)	Improved metrics, higher design complexity	Yes (simulator)	7 8
Optimization-enhanced QSVM	Wisconsin Diagnostic	98	98	0.995	Efficient (0.14s sim)	Yes (simulator)	9

Direct Answer

Recent studies (2023–2026) show that hybrid quantum-classical models, especially those using trainable quantum kernels and custom feature maps, achieve recall, F1, and ROC-AUC values in the range of 90–98% on breast cancer datasets, often matching or slightly exceeding classical SVM baselines. These approaches are reproducible within 48 hours on Qiskit or PennyLane simulators, but face computational trade-offs (longer runtimes, scalability limits) and require dimensionality reduction. Testable hypotheses include: (1) hybrid quantum-classical models can outperform classical SVMs on high-dimensional breast cancer datasets when optimized; (2) integrating explainable AI improves clinical trust; (3) evolutionary optimization of quantum feature maps boosts quantum kernel performance, making them feasible for rapid experimentation on simulators 1 4 6 10.

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None.

Study Scope

- **Time Period:** 2023–2026
- **Disciplines:** Quantum machine learning, medical AI, oncology, computational optimization
- **Methods:** QSVM, trainable quantum kernels, hybrid quantum-classical models, classical SVM baselines, feature mapping, optimization algorithms, explainable AI, benchmarking on Qiskit/PennyLane

Assumptions & Limitations

- Most results are from noiseless simulators, not real quantum hardware [11] [12].
- Dimensionality reduction (e.g., PCA) is required due to limited qubit counts, causing information loss [6] [13].
- Reproducibility is challenged by differences in quantum software frameworks and simulator precision [14] [15].
- Clinical validation and external benchmarking on diverse datasets are limited [2] [16].
- Computational costs for quantum kernel methods are higher than classical SVMs, especially as dataset size increases [17] [18].

Suggested Further Research

- Validate quantum kernel models on real quantum hardware and multi-center clinical datasets.
- Develop standardized benchmarking protocols across Qiskit and PennyLane.
- Integrate advanced explainability techniques (e.g., SHAP, LIME) for clinical trust.
- Explore noise-resilient quantum algorithms and evolutionary optimization of feature maps.
- Investigate quantum advantage in imbalanced, multi-class, and high-dimensional breast cancer data.

1. Introduction

1.1 Background and Motivation

Breast cancer remains a leading cause of morbidity and mortality worldwide, with early and accurate diagnosis critical for improving patient outcomes. Traditional machine learning methods, particularly classical Support Vector Machines (SVMs), have been widely used for breast cancer classification, but their performance can plateau in complex, high-dimensional, or limited-data scenarios. Quantum machine learning, and specifically quantum kernel learning, offers the potential to encode data into high-dimensional Hilbert spaces, enabling more expressive feature representations and potentially superior classification performance [1] [5] [10] [11].

Quantum kernel methods, including Quantum Support Vector Machines (QSVMs), trainable quantum kernels, and hybrid quantum-classical models, are being actively explored for breast cancer classification, especially under constraints of limited training data and reduced features. The promise of quantum advantage is tempered by current hardware limitations, noise, and the need for dimensionality reduction, making reproducibility and practical feasibility key concerns [5] [10] [11] [19] [20].

1.2 Scope and Structure of the Report

This report synthesizes recent advances (2023–2026) in quantum kernel learning for breast cancer classification, focusing on:

- Comparative performance of QSVM, trainable quantum kernels, and hybrid models versus classical SVM baselines.
- Quantum advantages in recall, F1, and ROC-AUC, and computational trade-offs.
- Novel feature mapping and optimization strategies.
- Datasets, reproducibility, and feasibility of rapid implementation (48 hours) using Qiskit or PennyLane.
- Research gaps and testable hypotheses for future work.

2. Recent Advances in Quantum Kernel Learning for Breast Cancer Classification

2.1 Quantum Support Vector Machines (QSVM) and Fixed Quantum Kernels

QSVMs leverage quantum feature maps (e.g., ZZ-Feature Map) to encode classical data into quantum states, enabling kernel estimation in high-dimensional spaces. Studies applying QSVMs to breast cancer datasets (e.g., Wisconsin Diagnostic) report moderate to high accuracy (70–94%), with performance dependent on feature selection and dimensionality reduction [1] [18] [21] [22]. Fixed quantum kernels, however, often suffer from exponential concentration effects, limiting their discriminative power and scalability compared to classical kernels like RBF-SVM [18] [21].

2.2 Trainable Quantum Kernels and Variational Quantum Circuits

Trainable quantum kernels, including those based on variational quantum circuits (VQC) and data re-uploading, enhance model expressivity and classification performance. By optimizing quantum kernel parameters and employing sub-sampling during training, these methods maintain or improve accuracy while reducing computational costs [3] [4] [23] [24] [25]. Hybrid quantum-classical SVMs with trainable quantum kernels have achieved up to 98.25% accuracy on the Breast Cancer Wisconsin dataset, outperforming classical SVMs (94.74%) and pure quantum SVMs (93.86%) [4].

2.3 Hybrid Quantum-Classical Architectures

Hybrid models integrate quantum kernels with classical machine learning or deep learning components, leveraging classical preprocessing (e.g., PCA, feature selection) and quantum feature extraction. These architectures are particularly effective with small datasets and reduced features, achieving high accuracy (up to 98%) and AUC (~0.995) [2] [4] [16] [23] [26]. Examples include hybrid quantum-classical convolutional neural networks (QCNN), quantum-enhanced neural networks (QENN), and ensemble models combining QSVM, VQC, and classical classifiers.

2.4 Novel Quantum Feature Mapping Techniques

Beyond standard maps like ZZ-Feature Map, recent studies propose quanvolutional layers, adaptive kernel selection frameworks, and custom-designed quantum feature maps. Quanvolutional layers act as quantum feature extractors integrated with classical neural networks, improving detection accuracy and addressing feature sparsity [25] [27] [28].

Adaptive frameworks select among multiple quantum feature maps (Angle, Amplitude, ZZ-entanglement) via meta-learning, optimizing kernel choice for breast cancer diagnostics [25].

2.5 Optimization Algorithms for Quantum Kernels

Optimization algorithms, such as elitist non-dominated sorting genetic algorithm (ENSGA), genetic algorithms, and quantum annealing, are increasingly integrated with QSVMs to enhance parameter tuning, classification accuracy, and computational efficiency [4] [9] [29]. These approaches automate feature map design, reduce training complexity, and improve generalization, supporting practical feasibility for rapid experimentation.

Synthesis

Recent advances demonstrate that hybrid quantum-classical models, trainable quantum kernels, and novel feature mapping techniques can achieve competitive or superior classification metrics compared to classical SVMs, especially under limited data and feature reduction. Optimization strategies further enhance performance and efficiency, making quantum kernel learning increasingly practical for breast cancer classification [4] [9] [10].

3. Performance Comparison: Quantum vs. Classical SVM Baselines

3.1 Metric-Based Performance: Recall, F1, and ROC-AUC

Hybrid quantum-classical models, especially those combining QSVM and QNN, consistently achieve recall, F1, and ROC-AUC values in the range of 90–98% on breast cancer datasets, often matching or slightly exceeding classical SVM baselines [4] [8] [19] [20]. Custom quantum feature maps can improve ROC-AUC by up to 5% and F1-score by up to 18% compared to standard quantum kernels [8].

3.2 Computational Costs and Scalability

Quantum kernel methods generally incur higher computational costs and longer training times than classical SVMs, especially as dataset size and feature dimensionality increase [4] [9] [14] [17] [30]. Sub-sampling and optimization techniques help mitigate these costs, but scalability remains limited by hardware constraints and the need for dimensionality reduction.

3.3 Impact of Custom Quantum Feature Maps

Custom-designed quantum feature maps, often generated via evolutionary algorithms or neural auto-designers, outperform standard maps, enhancing classification accuracy, AUC, and F1-score [7] [8] [20] [31] [32]. These maps improve adaptability to specific datasets and noise resilience, supporting robust performance in complex medical classification tasks.

3.4 Hybrid QSVM-QNN Models: Balancing Accuracy and Efficiency

Hybrid QSVM-QNN models leverage quantum feature maps and variational quantum classifiers to achieve high accuracy (>90%) while reducing computational cost compared to classical models [2] [4] [5] [9] [26]. Integration with optimization algorithms (e.g., ENSGA) further enhances efficiency, making these models practical for rapid experimentation on simulators.

Synthesis

Quantum kernel methods, particularly in hybrid architectures, achieve competitive recall, F1, and ROC-AUC metrics compared to classical SVMs, with custom feature maps and optimization strategies driving performance gains. However, computational trade-offs and scalability challenges persist, necessitating further research and hardware improvements [4] [7] [9].

4. Datasets, Reproducibility, and Feasibility in Quantum Kernel Learning

4.1 Commonly Used Breast Cancer Datasets

The most frequently used datasets in quantum kernel learning studies include:

- **Breast Cancer Wisconsin (Diagnostic and Original):** Tabular data, binary classification, features related to tumor morphology [1] [6] [16] [27] [33].
- **METABRIC:** Clinical and genomic data, high-dimensional, used for benchmarking quantum models under dimensionality reduction [6].
- **BreastMNIST, BCDR:** Imaging datasets, used in quantum convolutional neural network studies [27].
- **Mammographic Mass (MM):** Used for explainability and interpretability studies [34].

4.2 Reproducibility on Qiskit and PennyLane

Reproducibility is challenged by differences in quantum software frameworks, simulator precision, and integration complexity. Cross-SDK benchmarking shows that Qiskit and PennyLane produce nearly identical predictions at double-precision levels, but runtime and accuracy can vary [12] [14] [15] [35]. Standardized benchmarks and transparent reporting are essential for consistent evaluation.

4.3 Feasibility of 48-Hour Implementation

Quantum kernel learning experiments are feasible within 48 hours on simulators (Qiskit/PennyLane), especially with dimensionality reduction and efficient optimization techniques (e.g., sub-sampling, ENSGA) [2] [3] [5] [23]. However, real-device deployment and clinical validation remain future challenges.

4.4 Dimensionality Reduction and Feature Selection

Principal Component Analysis (PCA) is the most commonly used dimensionality reduction method, often reducing features to a manageable number of principal components [6] [23] [33] [36]. Hybrid approaches combining classical feature selection (e.g., Random Forest, Boruta-GA) with quantum feature maps improve accuracy and interpretability.

Synthesis

Quantum kernel learning for breast cancer classification is reproducible and feasible for rapid experimentation on simulators, but faces challenges in hardware deployment, information loss from dimensionality reduction, and external validation. Standardized benchmarking and integration of advanced feature selection are critical for future progress [1] [2] [14].

5. Explainability and Interpretability in Quantum Kernel Models

5.1 Explainability Methods for Quantum Kernels

Explainable AI (XAI) techniques, including SHAP, LIME, and Q-MEDLEY, are increasingly integrated with quantum kernel models to enhance interpretability and clinical trust [2] [16] [25] [37]. These methods identify key features influencing predictions (e.g., texture_worst, concave points_worst) and provide insights into model decision processes.

5.2 Clinical Relevance and Trustworthiness

Interpretability is essential for clinical adoption, enabling practitioners to understand and trust quantum kernel models. Integration of XAI techniques with quantum frameworks improves transparency, supports biomarker

identification, and enhances diagnostic decision support [16] [38] [39] [40].

Synthesis

Explainability and interpretability are underexplored but critical for clinical translation of quantum kernel models. Integration of XAI techniques enhances transparency and trust, supporting adoption in breast cancer diagnostics [16].

6. Research Gaps and Future Hypotheses

6.1 Hardware and Scalability Limitations

Quantum kernel learning is limited by hardware constraints (qubit count, noise, circuit depth), reliance on simulators, and the need for dimensionality reduction, restricting scalability and real-world deployment [6] [11] [38] [41] [42].

6.2 Impact of Dimensionality Reduction

Information loss from PCA and other feature reduction techniques negatively impacts quantum model performance compared to classical models using full features [13] [43] [44] [45].

6.3 Transition from Simulation to Real Hardware

Performance drops are observed when transitioning quantum kernel methods from simulators to real quantum devices due to noise and hardware limitations [1] [11] [42] [46].

6.4 Testable Hypotheses for Quantum Advantage

- Hybrid quantum-classical models can outperform classical SVMs on high-dimensional breast cancer datasets when optimized and validated on real quantum hardware.
- Incorporating XAI within quantum frameworks improves clinical diagnostic trust without compromising accuracy.
- Evolutionary optimization of quantum feature maps enhances quantum kernel performance and computational efficiency.
- Quantum-enhanced convolutional neural networks with quantum feature maps achieve superior performance on multi-modal breast cancer data compared to classical CNNs [2] [16] [47] [48] [49].

6.5 Directions for Future Research

- Validate quantum kernel models on real quantum hardware and multi-center clinical datasets.
- Develop standardized benchmarking protocols across Qiskit and PennyLane.
- Integrate advanced explainability techniques for clinical trust.
- Explore noise-resilient quantum algorithms and evolutionary optimization of feature maps.
- Investigate quantum advantage in imbalanced, multi-class, and high-dimensional breast cancer data [4] [38] [50] [51].

Synthesis

Significant research gaps remain in hardware validation, dimensionality reduction, explainability, and external benchmarking. Testable hypotheses and future directions focus on validating quantum advantage, improving interpretability, and enhancing scalability for clinical translation [2] [16] [50].

7. Conclusion

7.1 Summary of Key Insights

Hybrid quantum-classical models, trainable quantum kernels, and custom feature mapping techniques achieve competitive recall, F1, and ROC-AUC metrics compared to classical SVMs, with optimization strategies enhancing performance and efficiency. Quantum kernel learning is reproducible and feasible for rapid experimentation on simulators, but faces challenges in hardware deployment, information loss from dimensionality reduction, and external validation. Explainability and interpretability are critical for clinical adoption, and significant research gaps remain in hardware validation, benchmarking, and noise resilience [1] [4] [6] [10].

7.2 Outlook and Clinical Implications

Quantum kernel learning holds promise for advancing breast cancer diagnostics, offering potential advantages in complex, high-dimensional, and limited-data scenarios. Integration of explainable AI, standardized benchmarking, and validation on real quantum hardware are essential for clinical translation. Future research should prioritize noise-resilient algorithms, evolutionary optimization, and multi-center validation to fully realize quantum advantage in breast cancer classification [16] [47] [50].

---**: Advance noise-resilient algorithms and error correction strategies for practical deployment [27] [28].

- **Multi-center and Multi-class Evaluation**: Expand studies to diverse datasets and clinical settings to confirm quantum advantage and generalizability [1] [45].

In summary, quantum kernel learning for breast cancer classification has made significant strides, particularly through hybrid quantum-classical frameworks, advanced feature mapping, and optimization techniques. While quantum models show promise in low-data, reduced-feature, and imbalanced scenarios, classical SVMs remain superior in scalability and efficiency. Feasible implementations within 48 hours are possible using Qiskit and PennyLane, but reproducibility and real-world validation are critical next steps. Future research should focus on real hardware testing, standardized benchmarking, explainable AI, and adaptive kernel selection to bridge current gaps and accelerate clinical translation.

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